Logics & Symbolic AI – cheat sheet

1 Propositional logic

- **Syntax** = manipulating symbols; **semantics** = meaning.
- Formula: symbol p, constants $T, \bot$; $(\neg F)$; $(A \bullet B)$.
- Syntactic/semantic: ¬/Not, ∧/And, ∨/Or, ⊃/⇒, ⊂/⇐, ≡/⇔, ↑/Nand, ↓/Nor.
- **Valuation** $v : F \to \{True, False\}$, consistent (4):
  1) $v(T) = True$ 2) $v(\bot) = False$ 3) $v(\neg F) =$ Not $v(F)$
  4) $v((A \bullet B)) = v(A) \blacksquare v(B)$.
- **Tautology**: $v(X) = True$ for every v.
- **Satisfiable** S: some v_0 makes all $X \in S$ True (SAT).
- X tautology $\iff \neg X$ not satisfiable.
- **Consequence** $S \models X$: every v satisfying S makes X True.
- **Key**: $S \models X \iff S \cup \{\neg X\}$ not satisfiable.
- Ex falso: $A, \neg A \in S \Rightarrow S \models X$ for all X.
- Monotony: $S \models X \Rightarrow S \cup \{A\} \models X$.
- Deduction: $S \cup \{X\} \models Y \iff S \models (X \supset Y)$.
- X tautology $\iff T \supset X$ tautology, since $v(T \supset X) = v(X)$.

2 CNF & resolution

- **Negation normal form**: ¬ only before symbols.
- $[X_1..X_n]$ = generalized disjunction ($v([]) = False$).
- $\langle X_1..X_n \rangle$ = generalized conjunction ($v(\langle\rangle) = True$).
- **CNF** $\langle C_1, .., C_n \rangle$, each **clause** $C_i = [\ldots]$.

	rewrite	negation
∧	$X \wedge Y$	$\neg X \vee \neg Y$
∨	$X \vee Y$	$\neg X \wedge \neg Y$
⊃	$\neg X \vee Y$	$X \wedge \neg Y$
⊂	$X \vee \neg Y$	$\neg X \wedge Y$
↑	$\neg X \vee \neg Y$	$X \wedge Y$
↓	$\neg X \wedge \neg Y$	$X \vee Y$
⊅	$X \wedge \neg Y$	$\neg X \vee Y$
⊄	$\neg X \wedge Y$	$X \vee \neg Y$

- Disjunction: $[..P..] \to [..A, B..]$.
- Conjunction: $[..P..] \to [..A..], [..B..]$.
- $\neg\neg P \to P$, $\neg T \to \bot$, $\neg\bot \to T$.
- **Resolution**: $[A, X, B] + [C, \neg X, D] \to [A, B, C, D]$.
- Proof of X: start $[\neg X]$, end with $[]$.
- For $S \vdash X$: start $[S_1], .., [S_n], [\neg X]$.
- X tautology $\iff X$ has a resolution proof.
- Prolog `A :- B` $= [A, \neg B]$.

Example $((A \supset B) \supset (A \supset C))$
$\langle [\neg(A \supset B), A \supset C] \rangle \to \langle [A \wedge \neg B, A \supset C] \rangle$
$\to \langle [A, \neg A, C], [\neg B, \neg A, C] \rangle \to \langle [\neg B, \neg A, C] \rangle$

- **Backward** chaining: from goal (resolution).
- **Forward** chaining: from assumptions (Hilbert, modus ponens $\frac{X \quad X \supset Y}{Y}$).
- $\models$ semantic (truth tables); $\vdash$ syntactic (proofs).
- **Sound**: $S \vdash F \Rightarrow S \models F$.
- **Complete**: $S \models F \Rightarrow S \vdash F$.

3 First-order logic

- **Terms**: variables, constants, functors $f(t_1..t_n)$.
- **Atomic formula**: predicate $p(t_1..t_n)$, arity n.
- Formulas: atomic, $\neg F$, $(F \circ G)$, $(\forall x)F$, $(\exists x)F$.
- **Functor or predicate?** Inside an argument ⇒ functor. At formula level (¬, ∧, ⊃, quantifier) ⇒ predicate.
- **Free** v **(4 rules)**: 1) occurs in atomic F 2) in $(\diamond v')F$ iff free in F and $v \neq v'$ 3) in $\neg F$ iff in F 4) in $F_1 \circ F_2$ iff in F_1 or F_2.
- **Closed** formula: no free variable.

Semantics

- Domain $D \neq \emptyset$; interpretation I (3): 1) $c^I \in D$ 2) $f^I : D^n \to D$ 3) p^I n-ary relation.
- Assignment A: $v^A \in D$. Model $M(D, I)$.
- **Attitude**: $p(t_1..)^{I,A} = True$ iff $(t_1^{I,A}, ..) \in p^I$.
- $(\forall x)F$: True for all B equal to A save for x; $\exists$: for one.
- **Valid**: true in every model (all A).
- F valid $\iff \{\neg F\}$ not satisfiable.
- Valid: $((\forall x)p(a,x)) \supset p(a,a)$.
- Not valid: $(\forall x)(p(a,x) \supset p(a,a))$, $(\forall x)(\exists y)p(x,y)$.

Prenex (rename variables first)

- $\neg(\exists v)A \equiv (\forall v)\neg A$; $\neg(\forall v)A \equiv (\exists v)\neg A$.
- $((Qv)A \wedge B) \equiv (Qv)(A \wedge B)$, either side, $Q \in \{\forall, \exists\}$.
- $((\forall v)A \supset B) \equiv (\exists v)(A \supset B)$ (flips!).
- $((\exists v)A \supset B) \equiv (\forall v)(A \supset B)$ (flips!).
- $(A \supset (Qv)B) \equiv (Qv)(A \supset B)$ (kept).

Skolemization

- $..(\exists x_k)..F \to F(x_k / f(x_1..x_{k-1}))$, new functor.
- Arity = number of ∀ before it (0 = constant).
- Preserves satisfiability; put ∃ first if you can.
- $(\forall x)(m(x) \supset (\exists y)s(x,y)) \to (\forall x)(m(x) \supset s(x, f(x)))$.

Prove F valid by resolution (6 steps)

1. $\neg F$ in prenex form (rename variables).
2. Skolemize; 3) drop ∀; 4) CNF (optional).
5. Resolution + **unification** $(P(x), \neg P(c))$: $x = c$).
6. $[] \Rightarrow \{\neg F\}$ unsatisfiable $\Rightarrow F$ valid.

- Propositional: sound, complete, **decidable**.
- FOL: sound, complete, **semi-decidable** (valid ⇒ proof found in finite time; not valid ⇒ may never stop).

4 Natural deduction

$$\text{Ax}\ \frac{}{\Gamma, p \vdash p} \qquad \neg\text{I}\ \frac{\Gamma, p \vdash \bot}{\Gamma \vdash \neg p}$$

$$\neg\text{E}\ \frac{\Gamma \vdash p \quad \Gamma \vdash \neg p}{\Gamma \vdash \bot} \qquad \text{Abs}\ \frac{\Gamma, \neg p \vdash \bot}{\Gamma \vdash p}$$

$$\bot\text{E}\ \frac{\Gamma \vdash \bot}{\Gamma \vdash p} \qquad \to\text{I}\ \frac{\Gamma, p \vdash q}{\Gamma \vdash p \to q}$$

$$\to\text{E}\ \frac{\Gamma \vdash p \to q \quad \Gamma \vdash p}{\Gamma \vdash q} \qquad \wedge\text{I}\ \frac{\Gamma \vdash p \quad \Gamma \vdash q}{\Gamma \vdash p \wedge q}$$

$$\wedge\text{Eg}\ \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash p} \qquad \wedge\text{Ed}\ \frac{\Gamma \vdash p \wedge q}{\Gamma \vdash q}$$

$$\vee\text{Ig}\ \frac{\Gamma \vdash p}{\Gamma \vdash p \vee q} \qquad \vee\text{Id}\ \frac{\Gamma \vdash q}{\Gamma \vdash p \vee q}$$

$$\vee\text{E}\ \frac{\Gamma \vdash p \vee q \quad \Gamma, p \vdash r \quad \Gamma, q \vdash r}{\Gamma \vdash r}$$

$$\forall\text{I}\ \frac{\Gamma \vdash F}{\Gamma \vdash \forall x F} \qquad \forall\text{E}\ \frac{\Gamma \vdash \forall x F(x)}{\Gamma \vdash F[t/x]}$$

$$\exists\text{I}\ \frac{\Gamma \vdash F(t)}{\Gamma \vdash \exists x F(x)} \qquad \exists\text{E}\ \frac{\Gamma \vdash \exists x F \quad \Gamma \vdash F \to G}{\Gamma \vdash G}$$

- ∀I: x not free in Γ. ∃E: x not free in Γ, G. Handout names: g = left, d = right.

Example $\vdash (A \to B) \to ((B \to C) \to (A \to C))$
$\Gamma = \{A \to B, B \to C, A\}$: Ax gives A, $A \to B$;
→E: B; →E with $B \to C$: C; then →I ×3.
Curry–Howard term: $\lambda f.\lambda g.\lambda x.g(fx)$.

5 Planning: STRIPS

- **State** describes the world; **actions** change it.
- Planning = action sequence from initial to goal state.
  Predicates/Variables Capitalized; constants lowercase.
- `Move(X,Y)` Pre: `At(X), Level(low)`
  Post: `!At(X), At(Y)`
- STRIPS solver: backtracking, semi-decidable.
- **Sussman**: goals `On(a,b)`, `On(b,c)` clash.

6 Planning as SAT (Kautz & Selman 1992)

1. Add **time** as the last argument: `Move(a,b,1)`.
2. **Ground** every predicate (all argument values).
3. Grounded clause → boolean: `move_a_b_12`, `U-1-2`.
4. Write the **(5 kinds)** of constraints below.

- 1) **Initial** ($t = 0$): every grounded fluent, + or ¬ (1 model).
- 2) **Goal** ($t = N$): only goal facts (many models).
- 3) **Problem**: e.g. table always clear, one action per step.
- 4) **Action**: $a_t \Rightarrow (p_t \wedge c_{t+1})$.
- 5) **Explanatory frame axioms (2 per fluent)**:
  – $(\neg f_t \wedge f_{t+1}) \Rightarrow \bigvee$ actions at t that add f.
  – $(f_t \wedge \neg f_{t+1}) \Rightarrow \bigvee$ actions at t that delete f.
- SAT is decidable but N is fixed: increase N until satisfiable.

7 Formal Concept Analysis

- **Context** $K = (X, Y, I)$, $I \subseteq X \times Y$ (objects × attributes).
- $A^\uparrow = \{y \in Y \mid \forall x \in A : (x,y) \in I\}$ (shared attributes).
- $B^\downarrow = \{x \in X \mid \forall y \in B : (x,y) \in I\}$ (objects having B).
- $\emptyset^\uparrow = Y$, $\emptyset^\downarrow = X$.
- **Concept** (A, B) **(2, both!)**: 1) $A^\uparrow = B$ 2) $B^\downarrow = A$. A extent, B intent.
- **Galois connection (4)**: 1) $A_1 \subseteq A_2 \Rightarrow A_2^\uparrow \subseteq A_1^\uparrow$
  2) $B_1 \subseteq B_2 \Rightarrow B_2^\downarrow \subseteq B_1^\downarrow$ 3) $A \subseteq A^{\uparrow\downarrow}$ 4) $B \subseteq B^{\downarrow\uparrow}$.
- **Closure operator (3)**: 1) extensive $A \subseteq T(A)$ 2) monotone 3) idempotent $T(T(A)) = T(A)$.
- $(\cdot)^{\uparrow\downarrow}$ closes extents; $(\cdot)^{\downarrow\uparrow}$ closes intents.
- Always a concept: $(B^\downarrow, B^{\downarrow\uparrow})$ and $(A^{\uparrow\downarrow}, A^\uparrow)$.
- **Order**: $(A_1, B_1) \preceq (A_2, B_2) \iff A_1 \subseteq A_2 \iff B_2 \subseteq B_1$.
- **Meet** $\left(\bigcap X_t, (\bigcup Y_t)^{\downarrow\uparrow}\right)$; **Join** $\left((\bigcup X_t)^{\uparrow\downarrow}, \bigcap Y_t\right)$.

Intersection method, row-based (Wille 1982; Stumme tutorial). Rows = objects = *transactions*: intersecting the item sets of transactions gives the closed itemsets = intents. Theorem: every intent is an intersection of row intents $\{x\}^\uparrow$ (dually: columns). **(5 steps)**

1. Top to bottom: intersect each row intent $\{x\}^\uparrow$ with every row above it.
2. Result not yet a row ⇒ append it as a new row; continue until the last added row.
3. Add a full row (Y) if none. Now each row = one intent; its extent = intent$^\downarrow$.
4. Draw: fewest crosses on top; link each circle to the circles whose row it contains (immediate only).
5. Labels: attribute y at the highest circle containing it ($\{y\}^\downarrow$); object x at the lowest ($\{x\}^\uparrow$). Intent = labels above, extent = labels below.

Check the drawing (4): 1) is it a lattice? 2) one upper neighbour ⇒ has an attribute label 3) one lower neighbour ⇒ has an object label 4) g below $m \iff (g,m) \in I$.

- **NextClosure**: intents in lectic order $y_1 < .. < y_n$; try $y_i \notin B$ from last to first:
  – $C = ((B \cap \{y_1..y_{i-1}\}) \cup \{y_i\})^{\downarrow\uparrow}$; accept if C adds no attribute before y_i.
  – Start $\emptyset^{\downarrow\uparrow}$; stop at Y. $O(|X||Y|^2|C|)$.
- **Clarified**: no identical rows/columns (isomorphic lattice).
- **Reduced**: no object/attribute derivable from others.
- **Implication** $Z_1 \Rightarrow Z_2$ valid $\iff Z_1^\downarrow \subseteq Z_2^\downarrow$.
- Counterexample: object in $Z_1^\downarrow \setminus Z_2^\downarrow$.
- **Armstrong (3)**: 1) reflexivity 2) augmentation 3) transitivity.
- **Basis (2)**: 1) complete 2) non-redundant.
  Duquenne–Guigues: pseudo-intents $P \Rightarrow P^{\downarrow\uparrow}$, minimal.

Slide example: table of concepts
Context: x_1:$y_1y_2y_3y_4$, x_2:$y_1y_3y_4$, x_3, x_4:$y_2y_3y_4$, x_5:y_1.

	extent A	intent B	below
C_1	$x_1x_2x_3x_4x_5$	$\emptyset$	(top)
C_2	$x_1x_2x_3x_4$	y_3y_4	C_1
C_3	$x_1x_3x_4$	$y_2y_3y_4$	C_2
C_4	$x_1x_2x_5$	y_1	C_1
C_5	x_1x_2	$y_1y_3y_4$	C_2, C_4
C_6	x_1	$y_1y_2y_3y_4$	C_3, C_5

NextClosure order: $C_1, C_2, C_3, C_4, C_5, C_6$. Clarify: $x_3 = x_4$, $y_3 = y_4$.

8 Association rules

- $\sigma(X)$ = number of transactions containing X.
- $s(X) = \sigma(X)/|T|$; frequent if $s(X) \geq minSup$.
- $s(X \Rightarrow Y) = \sigma(X \cup Y)/|T|$.
- $c(X \Rightarrow Y) = \sigma(X \cup Y)/\sigma(X)$.
- **Lift** $= c/s(Y) = \frac{s(X \cup Y)}{s(X)s(Y)}$: = 1 independent, > 1 positive.
- **Apriori**: support anti-monotone; infrequent ⇒ supersets infrequent.
- Pass k: join L_{k-1}, prune, scan, keep $\geq minSup$.
- Rules from L: $A \Rightarrow L \setminus A$, $c = s(L)/s(A)$.

FCA vs association rules (AR)

- FCA implication: deterministic, holds for all objects.
- AR: probabilistic (support, confidence, lift).
- Implication = AR with confidence 100%.
- FCA: small exact data. AR: large noisy data.

9 Fuzzy logic

- $\neg x = 1 - x$; Yager $\neg_w(x) = (1 - x^w)^{1/w}$.
- **T-norm (4)**: 1) commutative 2) monotone 3) associative 4) neutral $T(a,1) = a$. So $T(a,0) = 0$.
- **(4 common)**: min (largest); product ab; Łukasiewicz $\max(0, a+b-1)$; drastic (smallest).
- Weak contradiction: $T(a, 1-a) \leq 0.5$.
- **T-conorm** $\bot(a,b) = 1 - T(1-a, 1-b)$ **(4)**: commutative, monotone, associative, identity $\bot(a,0) = a$.
- max (smallest); $1 - (1-a)(1-b)$; $\min(a+b, 1)$; drastic (largest).
- **S-implication** $(\neg x \vee y)$: $\max(1-x, y)$, $1 - x(1-y)$, $\min(1 - x + y, 1)$.
- **R-implication** (residuum) $R_T(x,y) = \sup\{z : T(z,x) \leq y\}$:
  – $= 1$ if $x \leq y$;
  – else min: y; product: y/x; Łukasiewicz: $1 - x + y$.
- $I(\neg a) = 1 - a$, $I(a \wedge b) = T$, $I(a \vee b) = \bot$, $I(a \Rightarrow b) = R_T$.
- **Simple positive inference**: $P_1 \wedge .. \wedge P_n \Rightarrow_\phi C$.
  – Satisfied if $C \geq \phi(P_1..P_n)$.
  – Solution (3): 1) ≥ initial values 2) all rules satisfied 3) no output can be smaller.
  – Iterate rules in any order until all satisfied.
  – With negation: may have no solution / oscillate.
- **Fuzzy set** $\mu : U \to [0,1]$ **(3 sets)**: support $\mu > 0$; core $\mu = 1$; α-cut $\mu \geq \alpha$.
- α-capping $\min(\alpha, \mu(x))$.
- **Equality relation (3)**: 1) commutative 2) reflexive $E(x,x) = 1$ 3) transitive $T(E(x,y), E(y,z)) \leq E(x,z)$.
- Fuzzification $\mu(x) = \sup\{E(x,y) : y \in A\}$.
- **Defuzzification (3 methods)**:
  – First of Maxima $\min\{c \mid \mu(c) \max\}$: safe, ignores other maxima.
  – Mean of Maxima: may land on a non-maximum.
  – Median: equal mass on both sides; may also miss.
- **Mamdani (4 steps)**: 1) fuzzify inputs 2) $\alpha = T(\ldots)$ 3) cap C at α 4) defuzzify.

10 MAX SAT & PSL

- **Weighted MAX SAT**: world maximizing the total weight of satisfied formulas.
- Relaxation (Łukasiewicz, S-implication): clause degree $\min(\sum_i x_i, 1)$; negative literal gives $1-y(s)$.
- **PSL**: max $\sum_c w(c)\min(\sum_{s+} y(s)+\sum_{s-}(1-y(s)),1)$, $y\in[0,1]$, $y(s)\le v$.

Proof: Weighted MAX SAT is NP-hard (3 steps)
1. Take any SAT instance with clauses $C_1,\dots,C_m$.
2. Give every clause weight 1. Then: satisfiable $\iff$ optimum $=m$.
3. So a polynomial algorithm for Weighted MAX SAT decides SAT in polynomial time. SAT is NP-complete $\Rightarrow$ P = NP.

Conclusion: no polynomial algorithm unless P = NP; try all 2^n worlds. Decision version ("weight $\ge k$?") is in NP $\Rightarrow$ NP-complete.

Proof: PSL inference is polynomial (3 steps)
1. Each $\ell_c(y)=\sum_{s+} y(s)+\sum_{s-}(1-y(s))$ is affine in y.
2. $\min(\ell_c,1)$ is concave (min of affine); a sum with $w(c)\ge 0$ is concave; constraints are linear $\Rightarrow$ **convex** problem (no bad local optimum).
3. Same as the linear program (LP): max $\sum_c w(c)z_c$ s.t. $z_c\le \ell_c(y)$, $z_c\le 1$, $0\le y\le 1$, $y(s)\le v$. LPs are solvable in polynomial time (slides: hit and run).

11 Theorem proving & λ-calculus

- **Hilbert's program (3)**: 1) consistent 2) complete 3) decidable.
- **Gödel 1**: consistent system with arithmetic is incomplete.
- **Gödel 2**: it cannot prove its own consistency.
- Terms: variable x; abstraction $\lambda x.y$; application fx (left-associative).
- β: $(\lambda x.M)N \to M[N/x]$ (rename to avoid capture).
- **Strongly normalizing**: every reduction terminates.
- $(\lambda x.xx)(\lambda x.xx)$ never terminates; cannot be typed.

Church encoding (slides: number $\mapsto$ λ-term; below the λ-term of number m is written $\overline{m}$):

0	$\lambda f.\lambda x.x$	$n+1$	$\lambda n.\lambda f.\lambda x.f(nfx)$
1	$\lambda f.\lambda x.fx$	$m+n$	$\lambda m.\lambda n.\lambda f.\lambda x.mf(nfx)$
n	$\lambda f.\lambda x.f^n x$	mn	$\lambda m.\lambda n.\lambda f.\lambda x.m(nf)x$
True	$\lambda a.\lambda b.a$	$\neg$	$\lambda p.p$ False True
False	$\lambda a.\lambda b.b$	$\wedge;\vee$	$\lambda p.\lambda q.pqp$; $\lambda p.\lambda q.ppq$

- **Typed**: $\lambda x.y:\sigma\to\tau$; $fx:\tau$ needs $f:\sigma\to\tau$, $x:\sigma$.
- Simply typed λ is strongly normalizing (Curry).
- **Curry–Howard**: formula = type, proof = program.
  - $\to$**I** = abstraction; $\to$E (modus ponens) = application.
  - $\wedge$ = product, $\vee$ = sum, $\top$ = unit, $\bot$ = empty.
  - $\forall$ = dependent product, $\exists$ = dependent sum.
- Non-constructive proofs use excluded middle $P\vee\neg P$.

λ exercise sheet – solutions

Notation: $m,n=\lambda$-variables of the given term; $\overline{m}=$ the λ-term $\lambda f.\lambda x.f^m x$ of the *number* m, substituted for them.

Ex 1 *Give the reduced forms.*
- (1) $(\lambda x.x)y\to y$; (2) $(\lambda x.yx)z\to yz$.
- (3) $(\lambda x.\lambda y.x)(\lambda x.y)$: the free y would be captured $\Rightarrow$ rename λy to λw: $\to\lambda w.\lambda x.y$.
- (4) $(\lambda x.xy)(\lambda z.z)\to(\lambda z.z)y\to y$.
- (5) $(\lambda x.\lambda y.xy(xy))(\lambda z.zz)$ $\to\lambda y.(\lambda z.zz)y((\lambda z.zz)y)\to\lambda y.yy(yy)$.

Ex 2 *What does $\lambda n.n(\lambda x.false)true$ test about n?*
- Substitute $\overline{n}$: $\overline{n}(\lambda x.\text{false})\text{true}\to(\lambda x.\text{false})^n\text{true}$.
- $n=0$: true. $n\ge 1$: last step $(\lambda x.\text{false})(\dots)\to$ false. **Tests $n=0$.**

Ex 3 *What is $\lambda m.\lambda n.nm$?*
- $(\lambda m.\lambda n.nm)\overline{m}\,\overline{n}\to\overline{n}\,\overline{m}=\overline{m}$ composed n times.
- $\overline{m}(\overline{m}f)=f^{m\cdot m}\Rightarrow\overline{n}\,\overline{m}=\lambda f.\lambda x.f^{m^n}x=\overline{m^n}$: **power** m^n. Check $m{=}2,n{=}3$: f^8.

Ex 4 *Why is $\lambda n.\lambda f.\lambda x.n(\lambda g.\lambda h.h(gf))(\lambda u.x)(\lambda u.u)$ the predecessor?*
- Substitute $\overline{n}$; let $S=\lambda g.\lambda h.h(gf)$: $\overline{n}\,S(\lambda u.x)=S^n(\lambda u.x)$.
- $S(\lambda u.x)=\lambda h.h((\lambda u.x)f)=\lambda h.hx$: first f *swallowed*.
- After $k\ge 1$ steps: $\lambda h.h(f^{k-1}x)$; apply to $\lambda u.u$: $f^{n-1}x$.
- So $\lambda f.\lambda x.f^{n-1}x=\overline{n-1}$ (and $n=0$ gives $\overline{0}$).

12 Description logics

- Variable-free, decidable fragments of FOL.
- **KB = TBox** (concepts, roles) **+ ABox** (named individuals).
- $\mathscr{AL}$ **(6)**: $\top,\bot,\neg A$ (atomic only), $C\sqcap D$, $\forall R.C$, $\exists R.\top$.
- $\mathscr{U}$: $\sqcup$; $\mathscr{E}$: $\exists R.C$; $\mathscr{C}$: $\neg C$; $\mathscr{N}$: $\le nR$, $\ge nR$.
- $\mathscr{AL}\subseteq\mathscr{ALE}\subseteq\mathscr{ALC}\subseteq\mathscr{SHOIN}\subseteq\mathscr{SROIQ}$ (OWL 2 DL).

Semantics $\mathscr{I}=(\Delta^I,\cdot^I)$
- $A^I\subseteq\Delta^I$, $R^I\subseteq\Delta^I\times\Delta^I$.
- $(C\sqcap D)^I=C^I\cap D^I$, $(\neg C)^I=\Delta^I\setminus C^I$.
- $(\exists R.C)^I=\{x\mid\exists y,(x,y)\in R^I\wedge y\in C^I\}$.
- $(\forall R.C)^I=\{x\mid\forall y,(x,y)\in R^I\to y\in C^I\}$.

Translation τ_x to FOL
- $\tau_x(C)=C(x)$; $\sqcap\to\wedge$; $\sqcup\to\vee$; $\neg\to\neg$.
- $\tau_x(\exists r.C)=\exists y,r(x,y)\wedge\tau_y(C)$.
- $\tau_x(\forall r.C)=\forall y,r(x,y)\to\tau_y(C)$.
- $C\sqsubseteq D$: $\forall x(\tau_x(C)\to\tau_x(D))$.
- ABox: $C(a)$, $r(a,b)$.

Reasoning: TBox (4), ABox (4)
- TBox model: satisfies all $C\sqsubseteq D$, $C\equiv D$.
- Satisfiable: $C^I\neq\emptyset$ in some model of $\mathscr{T}$.
- Subsumption $\mathscr{T}\models C\sqsubseteq D$: $C^I\subseteq D^I$ in all models.
- Disjoint: $C^I\cap D^I=\emptyset$ in all models.
- $C\sqsubseteq D\iff C\sqcap\neg D$ unsatisfiable.
- Disjoint $\iff C\sqcap D$ unsatisfiable; C unsatisfiable $\iff C\sqsubseteq\bot$.
- ABox **consistent**: a model of $\mathscr{A}$ and $\mathscr{T}$ exists.
- **Instance**: $\mathscr{A}\models C(a)\iff\mathscr{A}\cup\{\neg C(a)\}$ inconsistent.
- **Retrieval**: all a with $C(a)$. **Realization**: most specific C.

Inconsistent ABox
- $\mathscr{T}$: Father $\sqsubseteq$ Man, Mother $\sqsubseteq$ Woman, Man $\sqcap$ Woman $\sqsubseteq\bot$.
- $\mathscr{A}$: Mother(mary), Father(mary).
- $mary^I\in Woman^I\cap Man^I=\emptyset$: inconsistent.

Ontologies & OWL
- Ontology **(4 words)**: 1) formal 2) explicit specification of a 3) shared 4) conceptualization.
- Properties **(5)**: inverse, transitive, symmetric, functional, inverse functional.
- Functional: $P(X,Y)\wedge P(X,Z)\Rightarrow Y=Z$.

13 DL exam-style exercises

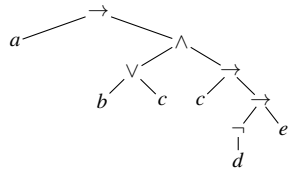
Prove $\forall r.(A\sqcap B)\sqsubseteq\forall r.A\sqcap\forall r.B$
- Take $x\in(\forall r.(A\sqcap B))^I$ and y with $(x,y)\in r^I$.
- Then $y\in A^I\cap B^I\subseteq A^I$, so $x\in(\forall r.A)^I$.
- Same for B, so x is in the intersection.
- FOL: $\forall x[\forall y(r\to A\wedge B)\to\forall y(r\to A)\wedge\forall y(r\to B)]$ is valid.
- The converse holds too ($\equiv$).
- $\exists r.(A\sqcap B)\sqsubseteq\exists r.A\sqcap\exists r.B$, *not* the converse.

Is Pedro a SuccessfulMan?
- SuccessfulMan $\equiv\neg$Female $\sqcap$ InBusiness $\sqcap\exists$married.Lawyer $\sqcap\exists$child.(StudiesAtUni $\sqcup$ Works).
- ABox: Pedro:¬Female, Pedro:InBusiness, Mary:Lawyer, John:Works, married(Pedro,Mary), child(Pedro,John).
- Yes: Mary witnesses ∃married; John witnesses ∃child.
- $\equiv$ is sufficient; with only $\sqsubseteq$ it would not follow.
- Method: check each conjunct; witness for each ∃; all fillers for each ∀.

14 Example exam – full solutions

Propositional Q1: when is $F=a\to((b\vee c)\wedge(c\to(\neg d\to e)))$ false?
- $X\to Y$ false iff $X=1$, $Y=0$: so $a=1$ and $(b\vee c)\wedge H=0$.
- Case 1: $b\vee c=0$: $b=c=0$, d,e free (4 rows).
- Case 2: $H=c\to(\neg d\to e)=0$: $c=1$, $\neg d=1$, $e=0$, so $d=e=0$, b free (2 rows).
- *abcde*: 10000, 10001, 10010, 10011, 10100, 11100 (6).

Propositional Q2: decomposition tree (root = main connective)



Propositional Q3: CNF
- $\neg d\to e\equiv d\vee e$; $c\to(d\vee e)\equiv\neg c\vee d\vee e$.
- $F\equiv\neg a\vee((b\vee c)\wedge(\neg c\vee d\vee e))$; distribute $\neg a$:
- $F\equiv(\neg a\vee b\vee c)\wedge(\neg a\vee\neg c\vee d\vee e)=\langle[\neg a,b,c],[\neg a,\neg c,d,e]\rangle$.

FOL Q1: function or predicate (and why)
- **Predicates** (operand of $\to,\neg,\vee,\wedge$, i.e. formula level): $a/2$, $b/3$, $d/1$, $e/1$.
- **Functions** (argument of another symbol, i.e. a term): $c/2$ in b, $f/2$ in e, $g/1$ in f.

FOL Q2: free variables
- $\forall x.\exists y.$ binds x,y only inside $(\neg b(\dots)\vee d(x))$.
- x,y in $a(x,y)$ and in $e(f(x,g(y)))$ are outside that scope: free.
- z is never quantified: free. Answer: $\{x,y,z\}$.

Planning ($pX1_3$ = `posX-1-3`, $L2_3$ = `L-2-3`)
- Q1 initial state: $pX1_0\wedge\neg pX2_0\wedge\neg pX3_0\wedge pY1_0\wedge\neg pY2_0\wedge\neg pY3_0$.
- Q2 goal (5 steps): $pX2_5\wedge pY3_5$.
- Q3 frame axioms, posX at $t=3$ (2 per value × 3 values = 6):
  - $(\neg pX1_3\wedge pX1_4)\to L2_3$; $(pX1_3\wedge\neg pX1_4)\to R1_3$
  - $(\neg pX2_3\wedge pX2_4)\to R1_3\vee L3_3$
  - $(pX2_3\wedge\neg pX2_4)\to R2_3\vee L2_3$
  - $(\neg pX3_3\wedge pX3_4)\to R2_3$; $(pX3_3\wedge\neg pX3_4)\to L3_3$
- Q4 Up at $t=2$ (one per Y value = 3):
  - $U1_2\to pY1_2\wedge\neg pY1_3\wedge pY2_3$
  - $U2_2\to pY2_2\wedge\neg pY2_3\wedge pY3_3$
  - $\neg U3_2$ (precondition $Y<3$ is false)

Fuzzy
- **Q1** $T(a,b)=\frac{a+b}{2}$ is **not** a T-norm: commutative, monotone hold; neutral fails $T(0.4,1)=0.7$; associativity fails $T(T(0,0),1)=.5\neq T(0,T(0,1))=.25$.
- **Q2** $R_{\min}(0.3,0.7)=1$ since $0.3\le 0.7$; $\vee$ = conorm of min = max: $\max(1,0.1)=\mathbf{1}$.
- **Q3** Rules: $c\ge\min(a,b)$, $a\ge c$. Start $a=0$ (no initial value), $b=.3$, $c=.1$.
  - Rule 1: $\min(0,.3)=0\le.1$ OK. Rule 2: raise a to .1.
  - Rule 1: $\min(.1,.3)=.1\le.1$ OK. Stable: $\mathbf{a}=.\mathbf{1},\mathbf{b}=.\mathbf{3},\mathbf{c}=.\mathbf{1}$.
- **Q4** First of Maxima, $d(\mu)=\min\{c\mid\mu(c)\text{ maximal}\}$:
  - $+$ always returns a point of maximal membership (safe); simple.
  - $-$ ignores the other maxima (best years 1960–2020 → only 1960); ignores the shape.
- **Q5** No (unless P = NP): reduce SAT (proof in section 10).
- **Q6** Yes: with $[0,1]$ values it is PSL, a convex problem (LP), polynomial.
- **Q7** Objective $f=a+2\min(2-a-b,1)+3b$:
  - if $a+b\le 1$: $f=a+2+3b$, best on $a+b=1$: $f=5-2a$;
  - if $a+b\ge 1$: $f=4-a+b$, maximal at $a=0,b=1$;
  - $\Rightarrow\mathbf{a}=\mathbf{0},\mathbf{b}=\mathbf{1}$, $f=0+2+3=5$.

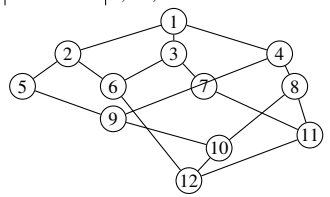
FCA: names → symbols
- x_1 Frodo, x_2 Aragorn, x_3 Gandalf, x_4 Legolas, x_5 Saruman, x_6 Boromir, x_7 Gollum.
- y_1 Middle-earth, y_2 Warrior, y_3 Magic, y_4 Immortal, y_5 Ring-carrier, y_6 Fellowship, y_7 Evil.

FCA Q1: clarification needed?
- **Yes**: rows $x_2=x_6\Rightarrow$ merge; no identical columns.
- y_1 is a full column (in every intent): reducible.

FCA Q2: lattice by the intersection method
- Rows: $x_1\{y_1y_5y_6\}$, $x_2{=}x_6\{y_1y_2y_6\}$, $x_3\{y_1y_2y_3y_4y_6\}$, $x_4\{y_1y_2y_4y_6\}$, $x_5\{y_1y_3y_4y_5y_7\}$, $x_7\{y_1y_5y_7\}$.
- New rows: $x_1\cap x_2=\{y_1y_6\}$, $x_1\cap x_5=\{y_1y_5\}$, $x_2\cap x_5=\{y_1\}$, $x_4\cap x_5=\{y_1y_4\}$, $x_3\cap x_5=\{y_1y_3y_4\}$; add full row $Y\Rightarrow$ 12 intents.

Table of concepts (every intent also contains y_1)

#	intent	extent	below
1	y_1	all 7	(top)
2	y_6	$x_1x_2x_3x_4x_6$	1
3	y_5	$x_1x_5x_7$	1
4	y_4	$x_3x_4x_5$	1
5	y_2y_6	$x_2x_3x_4x_6$	2
6	y_5y_6	x_1	2, 3
7	y_5y_7	x_5x_7	3
8	y_3y_4	x_3x_5	4
9	$y_2y_4y_6$	x_3x_4	4, 5
10	$y_2y_3y_4y_6$	x_3	8, 9
11	$y_3y_4y_5y_7$	x_5	7, 8
12	Y (all)	$\emptyset$	6, 10, 11



FCA Q3: closures ($^\downarrow$, $^\uparrow$, then $^\downarrow$ again gives the same extent)
- y_6: $\{y_6\}^\downarrow=\{x_1,x_2,x_3,x_4,x_6\}$; $^\uparrow=\{y_1,y_6\}$: concept 2.
- y_3: $\{y_3\}^\downarrow=\{x_3,x_5\}$; $^\uparrow=\{y_1,y_3,y_4\}$: concept 8.
- y_2: $\{y_2\}^\downarrow=\{x_2,x_3,x_4,x_6\}$; $^\uparrow=\{y_1,y_2,y_6\}$: concept 5.

FCA Q4: verify
- $(\{x_3,x_5\},\{y_3,y_4\})$: **no**, $\{x_3,x_5\}^\uparrow=\{y_1,y_3,y_4\}$.
- $(\{x_1,x_2\},\{y_6\})$: **no**, $\{y_6\}^\downarrow$ has 5 objects.

FCA Q5: implications ($|X|=7$)
- $\{y_2,y_6\}\to\{y_4\}$: $\{y_2,y_6\}^\downarrow=\{x_2,x_3,x_4,x_6\}$; x_2,x_6 lack $y_4\Rightarrow$ **invalid**; $s=2/7$, $c=2/4$.
- $\{y_5,y_6\}\to\{\neg y_7\}$: $\{y_5,y_6\}^\downarrow=\{x_1\}$, x_1 lacks $y_7\Rightarrow$ **holds**; $s=1/7$, $c=1$.

FCA Q6: association rules (AR) vs FCA
- Implication = AR with confidence 1: valid $\iff Z_1^\downarrow\subseteq Z_2^\downarrow\iff\sigma(Z_1\cup Z_2)=\sigma(Z_1)$.
- Concept intents = closed itemsets; $s(B)=|B^\downarrow|/|X|$; $s(Z)=s(Z^{\downarrow\uparrow})$, so the lattice holds all supports.
- FCA: small clean data, exact rules + hierarchy. AR: large noisy data, thresholds.
- Example: $\{y_2\}\Rightarrow\{y_6\}$ $c=4/4$ (implication); $\{y_6\}\Rightarrow\{y_2\}$ $c=4/5$, $s=4/7$ (AR only).

15 Proof templates

- **Galois**: $Y\subseteq A^\uparrow\iff A\subseteq Y^\downarrow$.
- **Antitone**: m shared by all of $X'\supseteq X$ is shared by X.
- $\alpha\beta\alpha=\alpha$: $\supseteq$ extensive; $\subseteq$ antitone on $X\subseteq X^{\uparrow\downarrow}$.
- **Meet is a concept**: $(Y_1\cup Y_2)^\downarrow=Y_1^\downarrow\cap Y_2^\downarrow=X_1\cap X_2$.